

AMAZON ECOMMERCE DATABASE MANAGEMENT\

CHAPTER-7

ORDER AND ORDER_DETAILS

Introduction – Orders and Order Details

The Orders and Order Details module is used to manage customer orders in an e-commerce database. The ORDERS table stores the main information about an order, such as customer ID, order date, and total amount. The ORDER_DETAILS table stores the individual products included in each order, along with their quantity and unit price. These tables are connected using ORDER_ID, allowing the system to efficiently manage customer product orders, update order information, and generate customer order history reports.

1.1ORDER TABLE:

ORDER_ID	CUSTOMER_ID	ORDER_DATE	TOTAL_AMOUNT
1	1	20-SEP-26	1200
2	2	20-SEP-26	850
3	3	21-SEP-26	1500
4	1	21-SEP-26	650
5	2	21-SEP-26	2200

1.2ORDER_DETAILS TABLE:

ORDER_DETAILS_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE
1	1	1	2	500

ORDER_DETAILS_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE
2	1	2	1	200
3	2	4	1	850
4	3	3	2	750
5	4	5	1	650

Table Relationships – Orders and Order Details

The ORDERS and ORDER_DETAILS tables have a one-to-many relationship through the ORDER_ID field.

- ORDER_ID is the Primary Key (PK) in the ORDERS table.
- ORDER_ID is the Foreign Key (FK) in the ORDER_DETAILS table.
- One order can contain multiple order details.
- Each order detail belongs to one specific order.
- PRODUCT_ID in ORDER_DETAILS identifies the product included in the order.

Relationship Diagram

ORDERS

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| **One Order**

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| **1 : N**

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ORDER_DETAILS

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| **Multiple Order Details**



Products in the Order

SAMPLE ORDER RECORDS:

ORDER_ID	CUSTOMER_ID	ORDER_DATE	TOTAL_AMOUNT
1	1	20-SEP-2026	1200.00
2	2	20-SEP-2026	850.00
3	3	21-SEP-2026	1500.00
4	1	21-SEP-2026	650.00
5	2	21-SEP-2026	2200.00

ORDER_DETAILS – Insertion Records:

ORDER_DETAIL_ID	ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE
1	1	1	2	500.00
2	1	2	1	200.00
3	2	4	1	850.00
4	3	3	2	750.00
5	4	5	1	650.00

2.MANAGE CUDTOMER PRODUCT ORDERS:

```
SQL> SELECT
  2     O.ORDER_ID,
  3     O.CUSTOMER_ID,
  4     O.ORDER_DATE,
  5     D.PRODUCT_ID,
  6     D.QUANTITY,
  7     D.UNIT_PRICE,
  8     O.TOTAL_AMOUNT
  9 FROM ORDERS O
 10 JOIN ORDER_DETAILS D
 11 ON O.ORDER_ID = D.ORDER_ID
 12 ORDER BY O.ORDER_ID;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_AMOUNT
1	1	20-SEP-26	1	2	500	1200
1	1	20-SEP-26	2	1	200	1200
2	2	20-SEP-26	4	1	850	850
3	3	21-SEP-26	3	2	750	1500
4	1	21-SEP-26	5	1	650	650

3.STORE ORDER DATE,QUANTITY,AND TOTAL AMOUNT:

```
SQL> SELECT
  2     O.ORDER_ID,
  3     O.ORDER_DATE,
  4     D.QUANTITY,
  5     O.TOTAL_AMOUNT
  6 FROM ORDERS O
  7 JOIN ORDER_DETAILS D
  8 ON O.ORDER_ID = D.ORDER_ID
  9 ORDER BY O.ORDER_ID;
```

ORDER_ID	ORDER_DAT	QUANTITY	TOTAL_AMOUNT
1	20-SEP-26	2	1200
1	20-SEP-26	1	1200
2	20-SEP-26	1	850
3	21-SEP-26	2	1500
4	21-SEP-26	1	650

4. Perform order insertion and modification operations.

```

SQL> SELECT
  2     O.ORDER_ID,
  3     O.CUSTOMER_ID,
  4     O.ORDER_DATE,
  5     D.PRODUCT_ID,
  6     D.QUANTITY,
  7     D.UNIT_PRICE,
  8     O.TOTAL_AMOUNT
  9 FROM ORDERS O
 10 JOIN ORDER_DETAILS D
 11 ON O.ORDER_ID = D.ORDER_ID
 12 ORDER BY O.ORDER_ID;

```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_AMOUNT
1	1	20-SEP-26	1	2	500	1200
1	1	20-SEP-26	2	1	200	1200
2	2	20-SEP-26	4	1	850	850
3	3	21-SEP-26	3	2	750	1500
4	1	21-SEP-26	5	1	650	650
6	3	21-SEP-26	2	2	900	1800

6 rows selected.

5. Generate customer order history reports.

```

SQL> SELECT
  2     O.CUSTOMER_ID,
  3     O.ORDER_ID,
  4     O.ORDER_DATE,
  5     D.PRODUCT_ID,
  6     D.QUANTITY,
  7     D.UNIT_PRICE,
  8     O.TOTAL_AMOUNT
  9 FROM ORDERS O
 10 JOIN ORDER_DETAILS D
 11 ON O.ORDER_ID = D.ORDER_ID
 12 ORDER BY O.CUSTOMER_ID, O.ORDER_DATE;

```

CUSTOMER_ID	ORDER_ID	ORDER_DAT	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_AMOUNT
1	1	20-SEP-26	1	2	500	1200
1	1	20-SEP-26	2	1	200	1200
1	4	21-SEP-26	5	1	650	650
2	2	20-SEP-26	4	1	850	850
3	3	21-SEP-26	3	2	750	1500
3	6	21-SEP-26	2	2	900	1800

6 rows selected.

Conclusion

The **Orders and Order Details** module successfully manages customer product orders in the e-commerce database. The ORDERS and ORDER_DETAILS tables store important information such as order date, customer details, products, quantity, unit price, and total amount. The relationships between the tables help maintain data consistency and make it easier to perform order insertion, modification, and customer order history operations.